ActInf Livestream #030.0 ~ "How to count biological minds: symbiosis, the

Livestream | 1:11:26

it's life
okay
it is
hello and welcome everyone thank you to
joining the active inference lab
it is october 1st 2021 and
i'm daniel i'm here with blue and we're
going to be talking about this paper how
to count biological minds symbiosis the
free energy principle and reciprocal
multi-scale integration
welcome to the active inference lab
we are a participatory online lab that
is communicating learning and practicing
applied active inference you can find us
at the links here on this page
this is recorded in an archived live
stream so please provide us with
feedback so that we can be improving our
work
all backgrounds and perspectives are
welcome here
and we'll be following good video
etiquette for live streams
we are heading into october and the
number 30 paper about biological minds
we currently have the author matt sims
scheduled to come on the october 5th
discussion and then we'll have the 30.2
another participatory group
discussion so check out the calendar
here as well as these other series if
you want to
join or
suggest somebody who we can invite

today in acting stream number 30.0 the
goal is to learn and discuss
this paper how to count biological minds
and it's by matthew sims it's from 2020
in the journal synthese
and just like all the other zero videos
this is just an introduction to some of
the ideas some background that blue and
i whip together literally leading up to
the time of now so it's just an
introduction and if you want to
go deeper into an idea then come to the
discussion
we're going to go over the aims and
claims abstract roadmap of the paper and
then the figures and so on
so
let's just get started with the
introduction readings blue and what
makes you excited about this paper
you know daniel i'm always interested in
studying intelligence and how that
scales from
the subcellular to the systemic
ecosystem level
um
and so this is kind of an interesting
take and it echoes a lot of what we
talked about in
the discussion of mike levin's paper the
computational boundary of the self back
in july so i was excited that matthew
agreed to come on the live stream and
i've been excited about this paper um
just because it it takes active
inference
the multi-scale aspect of active
inference

in a really enhanced integrative
direction
how about you what did you like
i liked that it combined uh some general
ideas
and questions in philosophy philosophy
of biology and then tied it to the
natural history of a specific species
so it's all good you know make claims
about distributed systems or the
microbiome and the host in general but
then
tying it to a system that can be studied
and has a cool
ecological history is a good approach
kind of grounds the paper
here's the
paper
and how to count biological minds
symbiosis the free energy principle and
reciprocal multi-scale integration
i'll
read the first aim and then blue you can
say something or read the other ones
to support the notion of symbiotic minds
and the notion of nested biological
cognizers that falls out of it this
paper will develop and deploy the notion
of reciprocal multiscale integration
from within the free energy principle
multiscale integration describes the
case where each of two or more cognizing
systems uses the other to provide
evidence for its own model of the world
and itself acting in that world
as a case study in support of the notion
of symbiotic minds i will look at the
symbiosis of vibrio fischeri bacteria

and the bobtail squid
i will argue that the vibrio squid
assemblage constitutes a functionally
integrated cognitive hole the markov
blanket of which constrains those of
vibrio and the squid
and in showing how the notion of a
symbiotic mind can be supported by
reciprocal multi-scale integration
within the free energy principle
this paper contributes to the philosophy
of cognitive science
demonstrating that our folk
psychological conception of what cogniza
what cognizers are
require rethinking
moreover if the account of symbiotic
cognizance presented as tenable by
bringing into relief and accounted for
at least one possible transition in
cognition this paper contributes to a
more complete understanding of the
evolution of cognition
so cool claims and the only kind of two
halves to it of the general argument and
the specific
and
symbiosis and ecological association
have been
approached in
the free energy principle framework but
i think this reciprocal component
it's approached a little differently
here than probably in some works but
we'll get into that
okay
i'll do the first abstract one
the notion of a physiological individual

has been developed and applied in the
philosophy of biology to understand
symbiosis an understanding of which is
key to theorizing about the major
transitions in evolution from
multi-organismality to multicellularity
the paper begins by asking what such
symbiotic individuals can help to reveal
about a possible transition in the
evolution of cognition
it makes me think about the transition
from a solitary insect cognition to a
colony
so there's the change in the level of
organization and then is that purely
cognitive do cognitive changes preceded
do they follow what is the relationship
between the evolutionary the
physiological and the cognitive changes
is what the first part is getting it
the second part such a transition marks
the movement from cooperating individual
biological cognizers to functionally
integrated cognizing unit
somewhere along the way did such
cognizant units simultaneously have
cognizers as parts
expanding upon the multi-scale
integration view of the free energy
principle this paper develops an account
of reciprocal integration
demonstrating how some coupled
biological cognizing systems when
certain constraints are met can result
in a cognizant unit that is in ways
greater than the sum of its cognizing
parts um and yeah this paper just shouts
out a lot to seth mary or smith

mary a kid says last name and nader
smith and you know the major transitions
in evolution uh which is if you haven't
read that book or thought about those
transitions i mean you can even probably
pick up a summary um but it's really um
amazing to think about
the evolution in terms of the transition
from single cell to multi-cell and you
know from asexual to sexual reproduction
and there's you know i think seven major
transitions that happened in the course
of evolution and i think that this kind
of
is hinting at a cognizing evolution or
cognitive evolution which
we'll we'll get into a little bit later
hey parker
symbiosis between bee fishery
bacteria and the bobtail squid is used
to provide an illustration of this
account a novel manner of
conceptualizing biological cognizers as
gradients is then suggested
lastly it is argued that the reason why
the notion of ontologically nested
cognizers may be unintuitive stems from
the fact that our folk psychology notion
of what a cognizer is
has been deeply influenced by our folk
biological manner of understanding
biological individuals as units of
reproduction
so that's a really cool point because it
involves um
our understanding of biology models of
biology influence the way that we see
the world which is kind of

sounds by definition but it's really
interesting because people have thought
so many different things about different
parts of biology about animal biology
all of evolution
all of the things we see
so then how that
intersects with our understanding of
cognition and what cognitive levels
exist what levels of individuality exist
cool topics
so just a thought here
so what uh
a biological individual like a what a
cognizer is has been in uh
influenced by our understanding of
biological individuals as units of
reproduction so is a cognizing unit a
unit of cognition
it's like
is there a corresponding in
to cognitive evolution as um
corresponding units as there are to
biological evolution sorry
good
so those are the main pieces
the roadmap
not that many sub sections starts with
an introduction and goes through a nice
description of the free energy principle
which we're going to
use
in multiple slides here
there's a distinction drawn which we'll
also go into about mirror versus
adaptive active inference
multi-scale integration will be
discussed

and different kinds of multi-scale
integration come up later on
but then there's two sections about this
specific bacterial squid symbiosis
and then that symbiosis is returned to
with this
enhanced background on the
uni-directional and the reciprocal
multi-scale integration
so that's kind of the overall roadmap
keywords or anything to add on that blue
all right
keywords were
free energy principle active inference
symbiosis nested markov blankets
multi-scale integration cognitive
evolution emergence and physiological
individuals
so let's start with
symbiosis
so
the quote from the paper was
symbioses are cooperative heterospecific
associations
in which
each symbiont partner mutually benefits
from
eg gaining nourishment shelter etc from
the presence of the other partners
so heterospecific means different
species
and uh this is a graphic and also those
interesting the footnote that just said
we're kind of talking about the positive
mutualistic form of
co-living
not commensalist or parasitic forms of
co-living or co-existing but even just

looking at pairwise interactions before
more complex loops are
taken into account there's a whole game
theory space of interactions
so
what should our models
include cooperative interactions
adversarial interactions
uh which which one of these edges or
relationships
have what game theory what cognitive
theories
so it's a it's a cool approach to take
on symbiosis that respects all the ways
that different systems like a lycan
versus a
eukaryotic cell with a mitochondria all
these different
co-living
styles
so just a thought on that um
you know all of these co-living styles
are transactional whether it's you know
through a
voluntary transaction or just straight
theft
um these kind of associations
remind me of how we're we're starting to
scale into studying
cities and states and thinking about
debt right so like the debt that we have
to each other like i borrow this or or
you borrow that or i benefit you know
you in this way and you benefit me in
this way and
that scales from
the you know
very basic biological unit all the way

through
ecosystems so
okay physiological individuals
so we're gonna come back to this idea
of what's an individual but
one
wrote in is this quote so one way
philosophers have addressed this
question of individuality is by
conceptually developing and applying the
notion of a physiological individual
a biological individual of this kind is
a highly integrated functional unit
the heterogeneous parts of which
cohere together through regulatory
processes
like metabolic immunotolerance etc so as
to maintain the system's integrity and
resist environmental forces of decay
and the citation to prado 2011
looks at the
role of the immune system
and the interaction with the foreign
with the other during development
and uses that
relationship
to
explore the the notion of the interface
between the organism and the environment
and looks at like the physiological
story involving the microbe and the
social niche and all these other
components can come into play and that
that is in some
cases
a better explanation or prediction or
description
of a certain scenario such as cool

another cool topic what do you think
blue
so this this is cool the end of this
quote i reject the idea that there is no
possible distinction between the
organism and its environment i just want
to point that out
because we'll come back to that in uh in
two slides actually
yes it's a qualification of the
interface and the system but it's not
the rejection
that we just give up
so it's towards useful partitionings of
the system
okay
so
two models of individual individuality
more generally
there was a quote in the paper that said
the theoretical apparatus that falls out
of the free energy principle may be used
to arrive at an account of the kind of
enrichment of living processes that are
required for those processes to qualify
as cognitive processes ascribable to
physiological individuals so that's
linking the cognitive perspective
with the physiological perspective of
individuality
making good on the project of strong
life-mind continuity
so what models of individuality exist we
have physiological and cognitive
we have the um
view that is going to be implicitly
described as like folk biological which
is the darwinian replicator evolutionary

just a few other ones that exist there's
like the teleological concept of
individuality
which is um the end directedness
and the cons definition of the organism
as an end in it in and of itself
and that's when things are acting at the
organismal level
but there's probably a lot of distance
between what was that simplification and
what was written
the information theory of individuality
which i know blue you have something to
add on and then integrated information
theory
as well as pluralism uh and a sort of
historical perspective
that
looks at shifting boundaries of the
conception of individuality and then
tries to find some organization that
doesn't even use that as a central
divider i mean these are just some of
the ways but just also cool like you
know species but then there's so many
species concepts but individual seems
like that's the one that you can pick
out
but then it's not always that
way well and i guess we'll come back to
that
remark when we uh talk about markov
blankets here in a little bit
um so
the information theory of individuality
i absolutely love love love love this
paper um because for many reasons but
but i just wanted to highlight um in

this paper they talk about how
individuality can be nested and it
requires a bi-directional information
flow which which i think that this
distinction um is important right so
there has to be like control from the
top down and from the bottom up so
information has to go both ways and that
is what describes
an individual so
you know if you have um
i'm trying to think of like a
non-individual
like like okay you can have someone
broadcasting instructions through a
megaphone you know right like like
shouting you know like please move to
the right please move to the right like
so that that's like top down control but
there's no feedback from the crowd like
the right is blocked there's a path here
or there's you know a gap here we can't
walk through it or whatever if there's
no you know feedback then that's not an
individual right so you have this you
know top down control but nothing comes
back so in an individual there's always
that that reciprocal uh flow
and then the the forms of individuality
that they highlight in this paper are an
organism which is like you and me uh a
colonial form of
uh individual which is like a biofilm
right and then a driven form and this is
the part that goes back to that quote
about i reject the idea that you know
there can be no step or that
an organism is not distinct from its

environment and so in this enviro this
is environmentally driven um and that's
this last form of individuality that
they highlight so it's aware like the
nature of the individual is entirely
dependent upon the environment and
allowing for this type of individual to
exist
really allows the formation of life in
the very beginning because if you just
have collections of molecules
you know without a boundary or a barrier
that's environmentally driven and so
without that uh
organism being able to arise from its
environment you don't get the origin of
life and so i just wanted to just
highlight that
nice great points
the next key word is the free energy
principle or fep and this is from
active livestream 14 from the illusion
that carl firsten made to the wood lice
and his kind of
observations of nature
leading eventually
to
an integrative approach
that includes some of these areas like
considering how multi-scale systems work
and how collectives work so how things
work laterally within a level like
across and estimates and then multiscale
like the colony inside of the ecosystem
information and thermodynamics and
control theory and action selection as a
type of inference
we're going to

hear it here from the author and look
through the four main
points that they drew out of the fep
so this is again
coming from
the sims 2020 paper so read for more
detail
the free energy principle
starts from a particular view of life
that is grounded in statistical physics
so that's
an introductory claim an interesting way
to start
and again just pulling a few highlights
out to to get at this first key point
a biological system is one that self
organizes to a limited set of attracting
states that is far from thermodynamic
equilibrium so there it's like our body
temperature is kept above the room
temperature
far from the sort of naive chemical
equilibrium
and that's being maintained despite
dissipation being a prevalent in the
universe
and that
resistance to falling to equilibrium is
called the non-equilibrium steady state
density
importantly the nest density
towards which an organism's dynamics
flow corresponds to its phenotype i.e
its regular patterns of behavior
morphology and physiology to find itself
in its characteristic or phenotypic
states provides the organism evidence
that its behavior is countering

dispersive effects of random
fluctuations
so
even static seeming phenotypes like the
length of your femur it's actively
maintained by
cells being renewed and the turnover
rate and so on
so sometimes it's easier to think about
some phenotypes like cognitive
phenotypes as actually embodying
parametric values that are being kept
away from equilibrium
other times for more static components
of phenotype it's not that way but it's
just a cool way to think about phenotype
as just these measurements that are
stable over some time frame about the
world and the persistence of those
measurements through time
represents
the implementation of some strategy that
resisted dissipation
hence the maintenance of a
non-equilibrium steady state through
some kind of strategy
anything that
blue no
the second
the right the this brings us to the
second primary feature of fep
the markov blanket formalism
this is used in the context of fep to
describe a particular kind of
statistical organization of a system
relative to that which the system is not
namely it describes the statistical
partitioning of internal states and then

using just the variable names as they
are here
internal states are
the external states phi
and the sensory states s active states a
these latter states being
in the markov blanket which are the
sensory and the active states
so what what about this blue
just i mean we've talked um well on the
live stream a lot about markov blankets
and how they can be used to describe um
or or individual individuals so really
or to delineate the boundaries of an
individual and technically the the
markup blanket
defines conditional independence
so like
under what settings or under what
settings are these two systems
conditionally independent
yes
so there's some interface that is being
used and there's a conditional
independence across either side but yeah
cool a topic that's come up many times
and i'm sure this is not the end of it
this brings us to a third essential
feature of fep
the notion of a generative model
a generative model is a probabilistic
model that describes how the evolution
of sensory states of a markov blanket
could be caused by external states
it captures prior beliefs in the form of
probability distributions about
unobserved external states
and a likelihood mapping external states

to the evolution of sensory states
such models are quote implicit in the
dynamics of internal states
palacios at all 2020.
so
the mapping between
external states and the evolution of
estimates of internal states is this a
matrix
that is connecting the observations
being made through time
to the s matrix so that's a
bi-directional relationship with the
generative and the recognition model
that's the tail of two densities
but that's the signal processing uh
sort of uh
sampling the external world
and then having an ongoing estimate of
world states which could be just
unobserved world states or causes in the
world then there's how those states
change their time b
how policy interacts with how those
states change through time pi
d the prior over the states
and then the decision making of pi with
the c g and the e so this is the
partially observable markov decision
process
the pomdp
framing and it's not the only way we can
look at it but this gets like many of
the key
variables down which is why we and
there's uncertainties on all of these as
we've found so
this just gets a lot of the key pieces

down
the generative model could have a
different structure though this is not
the only structure
okay
good
so here is from the paper
importantly uh
you want to read this one
sure
importantly a systems generative model
may be cast in terms of its
non-equilibrium steady state
to see how this is the case requires
understanding the notion of dual
information geometry of self-organizing
systems and he gives several citations
one of which i guess is this one yep
so this is just giving the first
description
uh not like first in literature just
like a first introductory description on
this dual information geometry
this dual aspect concerns the intrinsic
information geometry of the
probabilistic evolution of internal
states
so that's like the actual if there's a
million neurons and they can each take
10 states
that's this
the geometry of the internal states is
like that many neurons can take on that
many values that's the internal
true state space of the actual internal
states
and the separate
in

extrinsic information geometry of
probabilistic beliefs about external
states that are parameterized by
internal states
so that n neurons could be fitting a
binary model is the light switch up or
down
or it could be a three-state model is it
up down or not either or both
whatever it's specifying that is a lower
dimensionality
than
the actual state space so it's like
there's a lot of
decisions in the whole computer but then
there's this external
realized component that's a smaller
state space
the extrinsic information geometry we
call these intrinsic ie mechanical or
state-based
and extrinsic i.e markovian or
belief-based information geometries
and uh
wanja has come on just to discuss
several papers so definitely
look at the work by all of these authors
because it's like
interesting topic we could probably
learn more about that and i know there's
people who are
listening who know more about it
okay
here
is
those intrinsic and extrinsic
information geometries
being used in

i think the sims paper
but
uh
not 100 sure actually but we can see
either that or it's in this previous
fristen paper
so there's the probabilistic flow the
system states over time that's the
intrinsic geometry
and that's that yeah what i think this
is the first paper okay okay and here's
the extras so then
basically this is just a visual showing
the higher dimensionality of the actual
components of the system that can
project onto like just two
lower areas
okay
the fourth
part
of fep
uh in the sims paper this brings us to
the fourth primary feature of fep
active inference
um
i'll read the yellow and then
you can give a thought on where you
think acting fits in
fep proposes that living systems are
able to resist the second law of
thermodynamics in virtue of avoiding
sensory states which are deleterious and
actively bringing about those sensory
states which allow them to maintain
their structural and functional
integrity
it is by engaging in this coupled
process of active inference that an

organism minimizes not only current free
energy encountered but expected free
energy i.e the free energy that would
arise were a particular action policy
selected and followed
um so instead of giving a thought i'm
going to give
an author's thought
on this
just to read a quote from the paper the
reason for using the free energy
principle to investigate and argue for a
symbiotic mind is not only because the
quantity free energy provides a measure
of cognition across
spatiotemporal scales but because the
free energy
principle and its various corollaries
suggest a plausible criterion for
identifying biological cognizers across
various various spatiotemporal skills
so that's why they touch the fep um just
for that reason which i thought was nice
i'm going to ask a nice question from
the chat from stephen
wrote
is there a dual information geometry
between intrinsic and extrinsic states
of active inference
and another dual information geometry
between action stage estates and niche
spaces
can i give a first answer
please okay
i would say that
the total state space
which includes the variables for the
internal the blanket and the external

space
that's
the shape it is
and then there's going to be sub shapes
like you could only look at the blanket
and the internal states that's the
particular states
so that's partitioning the system of
interest away from the niche
or you could look at it in a in a four
way or
there's probably a lot of other ways but
it seems like the full model is the full
geometry and then there's going to be
some reduced geometries
and then stephen clarified i.e a dual
information geometry in relation to the
generative model and what it is modeling
and a dual information geometry between
what actions are happening
and the niche modification
i think that gets at also that sigma
mapping function in the decosta paper
with like okay after there's conditional
independence
of the kind that the markov blanket
partitioning provides there's still so
are some internal
state partitions in the world that
maybe do some prediction about or
anticipation about external states so
it's as if they're estimating external
states
let's talk to somebody who knows
information geometry
that's it i was going to say but my
guess would be depends on where you put
the mark on blanket but definitely i

think uh anybody out there who's
listening that knows about information
geometry i'm curious to take a deep dive
down that
path so
so i encourage you to get in touch and
contact us come talk to us about
information geometry because i would
like to learn some more
there was a distinction drawn in the
paper between adaptive and mere active
inference so i actually don't think that
this has come up
we've seen several distinctions like
instrumentalism and realism in terms of
the interpretation but this is i think
another
relevant distinction
fep falls short of making the claim that
all markov blankets draw a line around
cognizers
it has been suggested that what
determines whether a living markov
blanketed system is cognizer
is whether or not it is the kind of
thing which engages in adaptive active
inference
rather than mere active inference
kirchoff 2018
adaptive active inference requires that
a system autonomously engage in active
inference maximizing sensory evidence
for its own existence so there's been a
coupled oscillator model
as well as the uh flywheel governor
model another classic kind of mechanical
regulatory system
and uh

as well as the discussion well does a
rock have a markov blanket that type of
question
and so this is from the kirchoff 2018
paper just drawing the distinction
between mere active inference which is
like the ability to draw a partition
around a system at all to distinguish it
from its niche and its interfaces
and then
its generative model could be like
nil or boring or just a coin flip
to contrast that with adaptive active
inference which is actually the inactive
sensory motor affordance semantic
coupling i'm a strange loop type
system what do you think about that blue
so i wonder you know when we talk about
um
active inference and adaptive active
inference
adaptive active inference um
implies more than just
mental action or maybe mental action is
sufficient but but i think
i feel like adaptive active inference
implies that there's more of an action
component and you know we've talked
about situations where where
you can have active inference occurring
without necessarily taking
an action or maybe a mental action um
you can actively infer what's happening
in the environment without
performing some kind of action that's
adaptive to it
i don't know maybe that's that's that
exactly it's action and non-action it

can be adaptive to not act so is that
mirror
waiting good question right
right right and also like you can have
adaptation that's occurring in the brain
like you adapt to a noise in the
environment like it's super loud and
sound super loud but then eventually
your brain adapts so are you still is
that mirror or is that adaptive
nice so
emergence
do you want to read whatever or say
whatever you'd like to say about
immersions
sure so um this is just the wikipedia
definition so i'll just read it um
emergence in philosophy systems theory
science an art emergence occurs when an
entity is observed to have properties
its parts do not have on their own
properties or behaviors which emerge
only when the parts interact in a wider
hole
and i just
included these
references
so so that that definition kind of
echoes integrated information theory of
consciousness which
again implies that like consciousness is
this emerging property when the parts
are interacting and create something
that's bigger than the whole and then um
both of these um wait oh yeah so both of
these papers um the roses paper and then
this paper by varley and eric hole
start to get into the mathematics of

emergence and they use similar
metrics it's like they use the partial
information decomposition over the the
five metric that's used in the
integrated information theory of
consciousness to kind of quantify
emergence like what is this
how much is this emergent property or
it's what degree is this an emergent
property of these parts um and then what
this is a strong and weak emergence um
paper that you you included daniel oh
yep so i just i saw it it was really
cool that first that you
chose some papers from people in the
active inference community and that
their recent papers like within the last
year or two
so just shows that there's development
on these topics and that's interesting
to think about and then i just thought
to throw in a third like two truths and
a lie but like
two newer citations and one older one
just the qualitative distinction between
strong and weak emergence
was influential by mark badao from 1997
and
that paper says i conclude that the
scientific and philosophical prospects
for weak emergence are bright and so
it's easy to be like oh i want like the
best version of whichever thing but in
the paper it distinguishes between
strong emergence as being sort of the
metaphysical discussion
about
how holes take on meanings

greater than the sum of their parts and
sort of all versus none like snapping
into existence in this kind of
semi-magical way
versus weak emergence like the ant
colony the city the actual complex
systems that can be modeled and so that
kind of operationalized emergence while
allowing a channel of
metaphysical discussion like you just
mentioned consciousness and other people
bring up emergent properties
consciousness all the time like
integrated information theory it's an
integrated information theory of
consciousness so
it's kind of just interesting how
there's like a metaphysical thread
in what this integration means
as well as a uh operational and a more
empirical side of this
discussion
and i also saw a recent 2021 paper i
can't remember the author but um dealing
with fmri and emergence like
in fmri which is kind of interesting i i
didn't i didn't include it but um i
thought that that was cool it was 20 21.
okay um
that's the mark of blankets go for it
that's it so again this camp this comes
from that uh here cough paper 2018.
um and you know this this figure is
called blankets all the way down um so
you know here they show a single-celled
organism and they predict what they they
illustrate one more conflict i model the
world um and then you have a multi-cell

organism a blanket of blankets that says
we modeled the world
um and then here it shows a pontiff that
shows blankets within blankets we model
ourselves modeling the world and that
was something uh interesting that david
krakauer actually said at the summer
institute um this last summer he said
you know humans is one of the things
that makes us
uniquely human is that we build models
of our model
we build models of our models um so i
thought that that was interesting
cool and then on the right side so
there's the uh
ppp example protozoa plants pontiffs
and then another
cyber physical way that the markov
blanket concept and the partitioning of
systems this way is being used is in
digital systems
so this is like
a team member and then there's
a certain type of blanket that they're
interfacing through in terms of sense
and action
and the database can also be thought of
as having different kinds of sensory and
action
states as well as internal generative
model it's dealing with a different
external set of states
that's from the vayatkin at all 2020
paper
how about
multi-scale integration you can go for
it

so i'll just read the yellow here um and
so this is from the ramstead owl paper
2019 which i think that that's where
this figure is from and then there's
also the hess but owl 2019 paper
it says the multi-scale integrationalist
view is a pluralist's theory about
cognitive boundaries it uses the notion
of nested markov blankets to demonstrate
that cognitive systems have a plurality
of ontological boundaries each relevant
to the study of cognition
according to this view each of the
spatio-temporally nested components of
this cognitive system may be
ontologically picked out by deploying
the markov blanket formalism at
different scales
and that's what we were talking about
earlier we talked about individuality at
different scales and we talked about the
markov blanket
individuating systems so
so this is just
one way that multi-scale integration can
be approached
but before the markow blanket even
was an idea there was other approaches
to partitioning multi-scale systems
so let's just go right to the author's
distinction in the paper
of what uni scale
integration is
and how it relates to the adaptive
versus mirror active inference
discussion and this is a question
stephen also raised in the chat
so

multi-skill integration so that's that
nesting of systems within one another
just
not i'm intentionally not adding any
adjectives just any kind of nesting
systems
multi-scale integration of living
systems that failed to possess
the high degree of autonomy required of
adaptive active inference
and hence fail to be cognizers continue
to engage in mere active inference at
fast time scale eg a mitochondria in a
cell
so that is visualized here before
returning to the quote like the circular
nodes on the bottom let's just say are
doing mere active inference they're
simple sensors or actuators
and then they sort of just in a very
clean way
lead to increasingly
weakly emergent behavior that starts
approximating adaptive active inference
so for example you have a python script
that is adaptive active inference just
so we don't have to even debate whether
you know snails or birds are doing it
just you write a script that is adaptive
active inference the sub functions
within it
even though they might have an input
output structure
and even a generative model
they're going to be a lot more
mechanical
or simple or linear not displaying the
properties of the actual active agent in

the python script at the higher level
so that's describing this process of
uniscale integration
and
then
just the paper describes more about how
adaptive active inference at the slower
scale of the cognizing system that's the
thought bubble here
drives the continued integration of the
nested markov blankets at scales below
it and not vice versa so the pythons
variables drive the patterns of the
computation of the lower level functions
but it does those don't meaningfully
feed back into higher level
properties
so adaptive active influence at the high
end
and then that's just being merely
explained mechanistically
by unidirectional integration with
systems that are decreasingly
cognizer-like
on some continuum of cognition
so i i like the the term that's used
here and i think i'll highlight why when
we come to um the nested cone slide
but he says the behavior of constituent
nested markov blanket systems are
enslaved to the slower dynamics of the
autonomizing cognizing organism so i
like this concept of enslaving
um the sub routines
okay
cognitive evolution do you wanna say
anything
uh sure so this is um

just i'll just read this little excerpt
from the people paper because the author
didn't technically use the term
cognitive evolution and i thought that
it might um refer to a couple different
things but uh i'll read this quote
because i think it's maybe highlights
the the point that the author was
getting at so this is a question about
the kinds of physiological individuals
that we can reasonably ascribe the term
cognizer to it's a question about how to
count biological minds the significance
of this question lies in the fact that
its answer may be used to shed light
upon possible major transitions in
cognition perhaps there are many such
transitions the move from reflex
behavior to sensory motor coordination
the move from non-sentience to sentience
to move from
individual intentionality to group
intentionality which this again echoes
the the major transition of of evolution
um so cognitive evolution i thought
about it in a couple ways and we're
coming to the cone slide next so maybe
we'll make a flow forward
so
you know i thought that that's one way
that we can think about cognitive
evolution from like the transitions in
evolution from say multicellular to or
single cell to multicellular um but here
this is from um mike levin's paper the
computational boundary of the self and
you can see this is like
the the cognitive um cone right so so

this is the cognitive cone of an
individual that the like cones that were
you know inverted and used in this way
but you can see that the the cognitive
cone of a tip
is much smaller than that of a dog is
much smaller than that of a human and so
perhaps this cognitive evolution is um
you know in terms of of
scale of of the organism or or
you know cognitive capacity of an
organism
but mike here uses this uh idea of
nested kind of this multi-scale nested
cognition
and here it's
shown here the cell the organism the
colony to represent these
compound intelligences and when we
talked on the stream back in july like
we talked about the warping of the um
the the cone that like the bigger cone
but like so the colony enslaves the
behavior of the organism enslaves the
behavior of the cell and so this goes
all the way down and that's why i liked
um what uh matthew sims his term
enslaving um in these nested more
complicated structures um because and
the idea that like if you were to grab
and twist the top of this cone you would
grab and twist all the cones in it
depending on how tight you twisted right
so so i like this idea of enslaving the
behavior and this was a good visual to
kind of tie that all together
that terminology also arises from
synergetics from hakken not fuller as

they say
um and this is footnote 10 an order
parameter is a notion taken from
synergetics and used in
dynamic systems theory
it denotes a measure of a global
system's macro scale unstable slow
dynamics that's like the sleep wake
that enslaves the fast dynamics of
microscale component systems that's like
the eeg rhythm and results in a globally
emergent pattern so it's about
multi-scale systems organizing but
synergetics
hackin not fuller approaches it from a
dynamic systems fuller approaches it
from a geometric perspective so there's
other ways to do this partitioning mark
how blanket is not the only partitioning
out there
but this was cool about
also
um levin's partitioning which we just
got to learn about
all right on to the squid
so i i don't know if these are the exact
right species but
one of the key uh
empirical examples of the paper the one
that's uh followed up on do you want to
describe the system
um
sure well so i mean the vibrio fischeri
um are the bacteria that colonize the
squid and yes it's the hawaiian bobtail
squid and shout out to michelle
nishiguchi and her lab that like was the
lab underneath my lab when i was in

biology lab so the juvenile squid um
recruits the vibrio fisheries into the
what's called a light organ and that the
bacteria are bioluminescent um and so it
becomes essentially like inoculated with
these um and so then it says i'll just
read this because it's easy it's
probably a better description so it's a
nocturnal predator of the shallow reef
um the berries and sands during the day
and hunts at night so it's always
whether or not it is preyed upon it's
always influenced with
this association with these
bioluminescent bacteria so the juvenile
is inoculated into the light organ by
through seawater
and then the host promotes the
colonization of uh vibrio fisheries and
only vibrio fisheries through the
production of mucus um which is bacteria
and food the elimination of competing
bacteria through hemocyte defenses of
its innate immune system and the
eventual shedding of the ciliated
dependent appendages and swelling of the
crypt membranes preventing further entry
into the light organ so the switch sucks
up the vibrio fisheries says we want you
we only want you
all of the other bacteria we're not
going to allow them like we're just not
going to facilitate that interaction um
and then it says following gradients
gradients of chitin which they feed upon
vibrio fischeri migrate deeper into the
crypts of the light organized organ
colonizing it and causing a biochemical

reaction resulting in their emission of
bioluminescent light
this occurs just in time for the
bobtails nightly comfort prey and
something else i thought that was
important for this um
cycle is that they're expelled every at
the every morning
right like so they just get out like so
so like they colonize the the vocal
split colonizes and sucks up all the
bacteria they multiply and divide inside
of the of the squid and then it shoots
them out so they're back in the sea
water and they're available for
recapture by other um
squid
nice so
we're going to return to
the markov blanket partitioning
and then
consider that
squid system and how
using the mark out blanket partitioning
we can distinguish between
uni-directional and reciprocal
multi-scale integration
so the nomenclature in this paper are
the internal states maybe because
there's r in the word
then s and a are the usual words for the
state
of sensory sense states coming in to r
and then active action states going out
of r influenced by so these are like the
set of isolating nodes
from which
knowing them you're knowing as much as

you can about r the internal states and
then here
phi or v i don't know which one it is is
the external states
okay
yes
okay
figure two is where we uh turn to these
two different kinds of multi-scale
integration
and directly compare them using
kind of the nomenclature and the
contribution i believe of this specific
paper
so to understand
the difference
uh i will deploy the terminology of
users you and resources are
users are living systems the internal
states of which inferentially generate
subpersonal predictions that allow them
to use other markov blanketed systems
that are external to them
so here is like the spider making its
web
so it's a user of this extended
cognitive resource
and so we can imagine that there's a
mark of blanket partitioning
that's around these entities
like we don't need to deny that there is
a partitioning between the web and the
spider
we can think about it using this u and r
maybe this unr is a partition we can ask
the author like is it a partitioning of
just markov
blankets from each other is there going

to be some sort of interface is that the
niche or
that's the unidirectional case
the bi-directional case
is where those two interacting systems
in the same way that the spider can be
thought of as the unidirectional user of
the resource of the web that the two
interactants
are sort of using and benefiting from
each other as an extended cognitive
resource
so that is the
one of the main points of the paper
looking at that difference between
kind of mere
multi-scale systems and then ones where
it's not just like a ladder it's almost
like a conversation
that is giving
some new emergence
outcomes
very cool okay
the next slide just shows some more
discussion on the
new
u and r relationships that arise
and then how the new non-equilibrium
steady state density arising from that
interaction
then
in the synergetic sense acts as a
ordering parameter for u1 and u2 so the
constituents are part of a new
ordering regime
because of the bi-directional
relationship
to be clear each constituent biocognizer

continues to engage in adaptive active
inference acting in ways that ensure it
remains statistically separate from its
environment
so again in the unidirectional case it's
mirror active inference on the bottom
that becomes increasingly cognitive as
you go towards some level of
autonomy like a pyramid scheme
this is like a different structure
where there's increasing levels of
autonomy of different kinds of
interacting systems
and then there's sort of a bigger
wrapper around that
okay
cool
it's like the the interaction creates a
new cone a new tent it builds a tent
yep makes sense
though then
uh
this slide
brings this emergence from the
functional because it's one thing to say
that the functional um you know these
two brain regions get coupled together
now there's a functional difference in
their firing rates that's kind of a low
mechanistic bar
the higher bar to pass is this concept
of ontological emergence like it really
becomes
a thing in and of itself
and
the authorized that's related to these
two ideas
that uh first that some properties of

the macro system cannot be reduced to
the structural properties of the
component parts and they're governing
microdynamics so that's sort of
anti-reductionism
or holism where that exists there's a
strong argument that you're kind of
losing something if you keep on
dissecting
and then two because of this irreducible
properties
a macro level system has ontological
status eg is an entity in its own right
so that just is a good question to think
about and to ask the author just which
systems are unidirectional integrators
which systems are
bi-directional
how do we know
why does it matter
those are just all
good questions to ask
so another question that i have is
to what degree
does do these new properties have to
exist like if we're going to quantify
emergence
in the very quantitative sense
is it any number greater than zero
or
where are we drawing the line at
what
number of emergent properties or what
quant what quantity of new properties
creates a new system
yeah it comes up with uh integrated
information theory and a lot of other
quantitative measures of

integration and consciousness because
it's like okay so if this system's one
and this one's a hundred thousand
are you gonna have now a second level
theory like 500 is the cutoff or is it
true that i could you know trade one of
these for 100 000 of these like once you
start putting everything on a numerical
value
and then that becomes your action
selection metric oh
if it's worth the square root of this
then it's worth this
so it sets up a market for action based
upon these metrics
which is pretty interesting that's a
bi-directional emergence
definitely figure three
is using the markov blanket partitioning
to look at the squid system
so uh
the squid is like user one resource two
so for species one it's the squid
bacteria system so resource species pair
one is the squid and it's first user and
second resource v fisheries u2 r1
so these show the two kinds of
interfaces like they're encircling each
other
so it's kind of like embedded one is
embedded within the other
from a causal
perspective they're the niche for each
other
because even though when you put them on
the two-dimensional map they're all just
like flat there on the plane
their nearest neighbors

are either themselves or the other one
so that's kind of a cool way to
expand the topology
of the just the histology of
the tissue
and then pull out like actually you know
each beta cell in the pancreas is
surrounded by other cells that aren't
beta cells something like that and the
other ones in a weird sense are also
surrounded by the other kind of cell
even though there's a lot more of this
non-beta cell type so just a cool way to
look at tissue interactions from the
mark all blanket perspective
you want to describe for or what
sure
or what
so so here in this um figure it shows
the the squid cycle um which we have
talked about uh we just we just kind of
talked about it so um
it shows at time zero
the squid is resting after venting um so
there's some
vibrio fishery left inside of the squid
they're not like you know sterilizing
themselves and so there's this is the
repopulation
of um vibrio fischeri and so then after
12 hours there's a high density of um
vibrio fischeri
and then it shows this degree of
interest so on the y-axis degree of
integration is the degree of
biocognitive individuality and so right
here when there's the most squid inside
vibrio fisheries um that's when there's

the peak of integration presumably
because the squid with you know three
vibrio fisheries inside of it is not
going to light up but when it
concentrates it has a whole lot of the
vibrio fischerite inside of the light
organ then that's when it's luminescent
um and that's when you know the novel
properties so like not the you know
well the you know vibration right alone
wouldn't be sufficient to be um like
there's not enough of them to be you
know luminescent
and then the split alone without the
vibrofisherie doesn't light up and then
it's not capable of of predation in the
way that it normally does
and so it's the peak of integration
happens right before the sun comes up
and then at dawn the vibrio fisheries
are expelled
and then the bioluminescence is over so
that's that's kind of when there's no
um integration between the two or
limited or minimal integration between
the two species
thanks for the explanation so a few
things on this
it's like there's no single cut off this
dashed line it's kind of like
how many grains of sand in the heap
like one
is usually you know too few
but 100 maybe you're getting there
so
it's this continuum of
cognitive embeddedness
and of emergent function and there is an

actual time
that that is most functional and that's
just because the world has an actual
structure where there's a certain time
where there's the most light and it's
the darkest outside
and then there's a time where it's
ineffective so if we can discount it
when it's ineffective and low density
and we can say that this is like the
peak of that
natural variability when it's the most
bright
of the bioluminescence and the most dark
outside there's something
that's varying between those two levels
and so this was just a cool way to show
that that there's a smooth
increase and that it's a little bit like
model specific where
we choose to draw the line with the with
who's the biological cognizer
so this was a cool pattern and to
include the environmental variation in
the cognition so it's not like here's
when you're good at taking tests or not
it was like here's the environment as it
changes and then here's how the organism
is
this reminds me of like you know
building a nightlight out of fireflies
so i don't know in california do you
guys have fireflies there when i was a
little kid they were everywhere on the
east coast and so we would catch
fireflies but like how many fireflies
does it take to make a nightlight like
you can't have one fight well it depends

on the size of your jar
and how many can you catch like you
can't create a nightlight with one
firefly like there's just not enough
light but if you get like you know 20 in
there then you have a good like
nightlight
so true how many make a party
right
okay this is fun um
implications of the vibrio squid
symbiosis
so just a nice uh
argument by nature
approach that i'm sure we could learn
more about
that how do you go from an observation
in the natural world oh this bird you
know the one with this chromosome set
does this for this long ergo
this is possible or this is something
about nature in general
so there are three philosophical
implications that are exposed by this
striking association between the vibrio
and the squid
that are significant to the account of
symbiotic minds which follows
so the first one
is that the vibrio squid is a clear
example of a symbiotic physiological
individual
and that's distinguished uh from the
other
non-physiological individual systems
like collections of physiological
individuals you know if i eat food the
other person doesn't get the nutrients

but then you go okay but
one person eating does provide the
nutrients for another person so it's
going to be a continuum and there's all
that's like sort of the complexity of it
but it's a great example of integration
the like i like it and then the second
one is um physiological individuals or
matters of degree over time the same
living system can be more or less of a
symbiotic physiological individual so i
think that really takes
that um almost unsaid criticism of the
first point
and immediately builds on it to say like
yes it is a continuum
and
that's the whole interesting part
is
you can see how the system's
cognitive interfaces change through time
and that's the interesting story like
for humans certainly our ability to um
do something alone in a room is gonna
change through time when we're born it's
very low and then there's an age range
where you can do some task alone in a
room
and then eventually there isn't so it's
like that is something that varies
continuously just like there's the daily
rhythm there's a lively rhythm
to some cognitive elements of all
systems
so it's not reducing them or explaining
them away it's actually looking at that
variability and what it says about the
system

and then
the last philosophical implication may
be thrown into relief when asking the
critical question
what is it that produces the
bioluminescence the squid the bacteria
or the temporary assemblage of these
organisms
so
is that a thing that deserves a noun
of itself because we have nouns for
assemblages of other things
of one type
like the barrel of oil or something like
that but is there a unit and a noun or
are we left to describe this only with a
process
and then default implicitly back to the
darwinian replicator
oh the squid's just trying to reproduce
and the bacteria is doing competition so
they're the individuals ergo is a
tenuous relationship or it's an interra
interaction ecologically but it's not a
thing in and of itself because it's not
the replicator
so
those are great questions raised by the
author
so the questions that i maybe have for
the author um
so
the bioluminescence
is there in the vibrio fisheries
it's just
self-detectable by us
or presumably by the prey that the
vibrio fisheries hunting right so so i

think
um
it's again like it's a continuum it's
not uh um
there is bioluminescence or there's not
bioluminescence there's like more or
less bioluminescence or detectable
bioluminescence that occurs at you know
x lumens
so so what is the
um
the line there that is wrong
good question so here's one of the last
points in the
discussion today
it's again integrating cognitive and
functional and physiological
individuality together
and then um what is in the background of
you why are those
when we say individuality and we have to
qualify it by saying well i mean
informational or i mean physiological
what are we qualifying it
implicitly or explicitly against
um i think the materialist worldview
might suggest that the individual is
like just the the one
corpuscular entity
like the one thing that can be separated
out like the ant is the nest mate worker
now there's an argument for the ampy and
the colony and the nest made being a
tissue but people say oh i saw 10 ants
on the windowsill not like i saw a
thousandth of an ants on the windowsill
so there's i think a pure materialist
argument for just separability

and that having the more noun-like
characteristic
and then
the author is arguing that
um there's this folk biological
concept that sometimes prevents
thinking about physiological individuals
as
real
uh bona fide
individuals
and that's this like evolutionary
individualism
model
uh which can even have a collaborative
or optimistic bent to it but it still
draws lines around what levels of
individuality exist in the world because
they're the ones that most cleanly
resemble darwinian replicators so not
the systems that grow slowly outwards or
transfer through different media in the
same way
and then the argument is that fep
and its corollary active inference the
mirror and adaptive and the continuum
between like we talked about
offer an ideal program for bringing our
folk psychological intuitions in line
with the way nature has carved its own
joints over evolutionary time skills
so that's an interesting
claim
definitely
okay
last
slide
this was like a william blake

line that i thought of when i saw the
uni-directional
the bird a nest the spider a web
man friendship people
friendship so it's a great
description of kind of this modification
and i think
it's getting at this idea that there's
niche modification
in nature
and we do it
that's sort of the unambiguous part you
know we change our environment but then
we're also our own environment
so then there's some dynamics of like
culture and communication and being each
other's niche
that are very different than the the
bacteria changing the ph of a solution
it can go up it can go down
but it's still ph
but then there's this different
boundedness to
what can happen with humans working
together
so that just brought a close these few
questions like
where is the
multi-scale integration happening
and then
how how do we identify the
unidirectional components
and the reciprocal components and uh
great paper
so just a final thought and something
that i've thought about a lot um in
terms of of reading this paper because
it's this colonization of the squid by

this bacteria i think a lot about
us and our colonization i mean there are
more
non-human cells in the body than there
are human cells which is fascinating so
but we also have the ability to
manipulate our niche our internal niche
the niche of the microbes that reside
within us through taking antibiotics but
there are lots of other things
probiotics and not just antibiotics but
but just even what you eat and where you
go and how
whether you're breastfed or not or go
through vaginal birth or not and all
these things contribute to your own
inoculation or how you inoculate your
children and this is something that you
can do like literally you can change the
composition of your microbiota like
through your lifetime
so
is that niche modification is it
what is that exactly
i think it's one reason why
a um
not neutral or
um
unbiased but flexible
integration
model like being presented here
we can just say okay rather than
wondering whether it falls on one side
or the other of a binary classification
like yes it's co
um symbiotic or not
we just have a descriptive framework
first

and then you can think about the
implications later or separate them
somehow but
we hope that if you're listening in the
right time you'll join us for the 30.1
and 0.2
because
the author hopefully will join for at
least the first one and it's cool topics
and we can do a little bit like we did
in 29.2
let's think of some systems
are they unidirectional are they
reciprocal
what reciprocal systems could exist like
what does that mean so
sounds good yep thanks uh people who are
listening and steven for the questions
live thanks blue for collaborating on
the slides here
and thanks to the author too for
providing us such such great food for
thought
totally agreed
so thanks and see you later